STATISTICS: DATA ORGANIZATION AND PRESENTATION

There are two types of data:
Discrete data: this is counted data e.g. people; cars; animals and so on
Discrete data cannot accommodate decimals/ fractions
Continuous data: this is measured data e.g. distance; mass; weight; height; temperature; time; marks; volume and so on
Continuous data does accommodate decimals/ fractions

There are two types of classes:
Inclusive type classes such as:

0 – 4
5 – 9
10 – 14
} Discrete Type.

0 - 9
10 - 19
20 - 29
And so on

0 – 9.5
9.5 – 19.5
19.5 – 29.5
etc
} Cont./Excl. Type classes.

Exclusive type classes such as:

0 - 10
10 - 20
20 - 30
And so on

0 – 5
5 – 10
10 – 15
} Cont. Type ⇒ Exclusive Type classes

Raw data: data which has not been org.

From the following data:

12	36	40	28	19	10	10	16	21	19	86	46
20	19	67	45	33	55	26	37	30	20	51	53
30	20	36	41	42	20	22	23	24	25	98	49
26	27	29	28	30	16	25	33	38	30	40	29
31	17	24	26	37	27	11	27	26	22	13	16
12	36	40	28	19	10	10	16	21	19	75	76
20	19	77	45	33	45	26	37	30	20	76	24
30	20	36	41	42	20	22	23	24	25	15	34
26	27	29	28	30	16	25	33	38	30	15	26
31	17	24	26	37	27	11	27	26	22	48	24
57	22	42	32	18	19	38	20	61	21	82	24
49	18	35	25	16	90	29	16	38	65	51	41
40	40	35	57	46	21	80	93	35	35	51	24
29	81	18	49	82	42	18	29	40	38	29	38
71	46	18	18	68	55	55	61	44	20	28	39
36	23	29	39	36	26	19	27	43	19	26	15
23	24	37	35	35	19	18	33				

Required:
1) Prepare a frequency distribution table by the exclusive method (employ the Sturges Rule)

The Sturges rule for determining the number of classes (k) in a frequency distribution with total frequency N is

$$k=1+3.322\log N= 1+3.322\log 200= 8.644021646 \approx 9 \text{ classes}$$

Prefer ⇒ class with

k=1+3.322logN= 1+3.322log200= 8.644021646 ≈ 9 classes

Prefer ⇒ class with ⇒ a multiple of 5 ⇒ 5, 10, 15 - - - - -

Range: Max minus Min sets in your data= 98-10=88

Class width/ Class magnitude/ Class size = h = Range/k= 88/8.644021646
=10.18044651

Approximately class width of size 10

Min value: 10

Max value: 98

Class	Tally Marks	freq
10-20	~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ IIII ///	38
20-30	~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ////	69
30-40	~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ /	41
40-50	~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~	25
50-60	~~IIII~~ ////	9
60-70	~~IIII~~	5
70-80	~~IIII~~	5
80-90	~~IIII~~	5
90-100	///	3
Total		200

1 — /
2 — //
3 — ///
4 — ////
5 — ~~IIII~~
6 — ~~IIII~~ /
⋮
10 — ~~IIII~~ ~~IIII~~

Frequency distribution table:

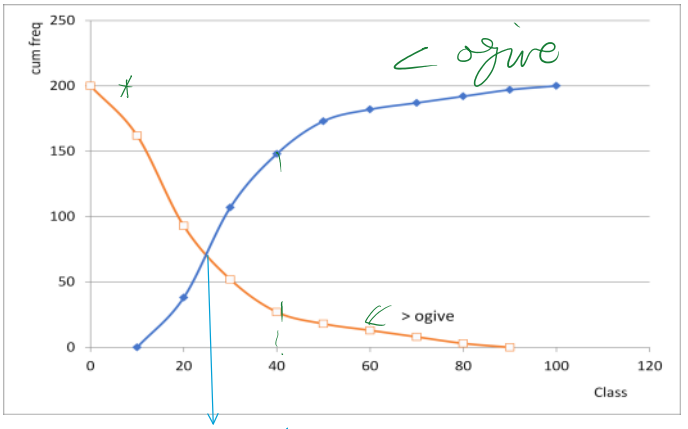
Class	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
freq	38	69	41	25	9	5	5	5	3

2) Represent the data with a less than and a more than ogive; hence estimate the interquartile range from the graph

Cum. freq. Curve

$Q.I. = \frac{Q_3 - Q_1}{2}$ → semi-Inter Q.R.

Class	F	<cf	>cf
10-20	38	38	200
20-30	69	107	162
30-40	41	148	93
40-50	25	173	52
50-60	9	182	27
60-70	5	187	18
70-80	5	192	13
80-90	5	197	8
90-100	3	200	3



Median

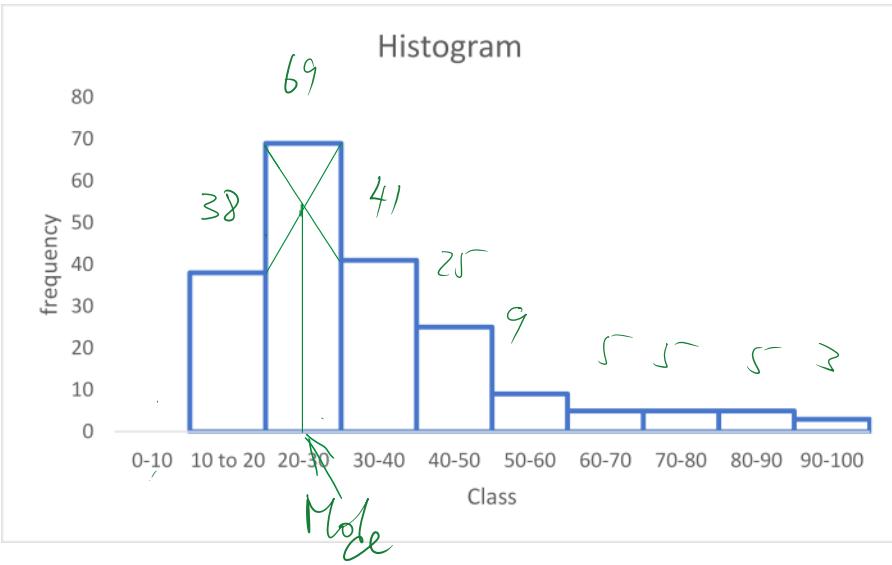
Inter-quartile Range = $Q_3 - Q_1$

To construct a less than ogive: plot the less than cumulative frequencies against the upper limit of the class intervals

To construct a more than ogive: plot the more than cumulative frequencies against the lower limit of the class intervals

At their point of intersection: if you drop an altitude, you obtain the median along the horizontal axis

Represent the data with a histogram and hence estimate the mode



$f_0 = 38$

$f_1 = 69$

$f_2 = 41$

Modal class = 20 − 30

$l = 20$

$h = 10$

Frequency distribution table:

Class	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
freq	38	69	41	25	9	5	5	5	3

Class	x	d	f	fd	fd^2	fd^3	fd^4	<cf
10-20	15	-4	38	-152	608	-2432	9728	38
20-30	25	-3	69	-207	621	-1863	5589	107
30-40	35	-2	41	-82	164	-328	656	148
40-50	45	-1	25	-25	25	25	25	173
50-60	55	0	9	0	0	0	0	182
60-70	65	1	5	5	5	5	5	187
70-80	75	2	5	10	20	40	80	192
80-90	85	3	5	15	45	135	405	197
90-100	95	4	3	12	48	192	768	200
			200	-424	1536	-4226	17256	

$$d = \frac{x - A}{} = \frac{x - 55}{10}$$

$$d = \frac{x - A}{h} = \frac{x - 55}{10}$$

Mean A=55; h=10

1. Calculate:

$$\bar{x} = A + \frac{h\sum fd}{\sum f} = 55 + \frac{10 \times (-424)}{200} = 33.8$$

Median

$$Md = l + \frac{h}{f}\left(\frac{n}{2} - c\right) = 20 + \frac{10}{69}(100 - 38) = 28.98550725$$

C ⇒ Cum. freq preceding the Md class

f_0 –

f_1 = highest.

f_2

Mode

$$M_o = l + \frac{h(f_1 - f_0)}{2f_1 - f_0 - f_2} = 20 + \frac{10 \times (69 - 38)}{2 \times 69 - 38 - 41} = 25.08474576$$

- **Standard deviation**

$$\sigma = \cancel{10}\,h\sqrt{\left(\frac{\sum fd^2}{\sum f} - \left(\frac{\sum fd}{\sum f}\right)^2\right)} = 10\sqrt{\left(\frac{1536}{200} - \left(\frac{-424}{200}\right)^2\right)} = 17.84824921$$

- **Lower quartile**

$$Q_1 = l + \frac{h}{f}\left(\frac{n}{4} - c\right) = 20 + \frac{10}{69}(50 - 38) = 21.73913043$$

- **Upper quartile**

$$Q_3 = l + \frac{h(\frac{3n}{4} - c)}{f} = 40 + \frac{10(150 - 148)}{25} = 40.8$$

- **Inter-quartile range**

$$Q_3 - Q_1 = 40.8 - 21.73913043 = 19.06086957$$

- **Quartile Deviation = (Q3 – Q1) / 2**

$$(Q_3 - Q_1)/2 = (40.8 - 21.73913043)/2 = (19.06086957)/2 = 9.530434785$$

↳ Semi-Interquartile Range

- **Coefficient of variation**

← is used to compare 2 sets of data!

$$CV = \frac{\sigma}{\bar{x}} \times 100\% = \frac{17.84824921}{33.8} \times 100\% = 52.80547104$$

NB/ Data with higher CV ⇒ more varied ⇒ heterogeneous
" " lesser " ⇒ more homogeneous
↳ less varied

Calculate the Karl Person's Coefficient of Skewness

$$S_k = \frac{3(Mean - Mediam)}{Std - Dev} = \frac{3(33.8 - 28.9855)}{17.8482} = 0.8092$$

Calculate Bowley's Measure of Skewness

$$S_b = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1} = \frac{40.8 + 21.7391 - 2 \times 28.9855}{40.8 - 21.7391} = 0.2397$$

Calculate Measure of Skewness based on moments

Calculate Karl Person's measure of Kurtosis based on moments